

The race engineer for the drivers who don't have one.

A professional race engineer costs hundreds of pounds per day. Every Formula 1 racers, veteran-team drivers, and grassroots competitors do not. APEX changes t

BUILT ON IBM GRANITE

ADAPTIVE RACERS FIRST

APEX DOT RACE

Regulatory barrier dropped in 2017. Economic barrier stayed.

The FIA lifted its single-seater ban on disabled drivers in December 2017. The barrier stopped being regulatory. A paid race engineer is a luxury good for amateur + clubman series.

Three constituencies, one shared product gap: adaptive racers, veteran-team drivers, and grassroots club race-engineer tools (Track Titan, Trophi.ai, Deep Dynamics) assume able-bodied physics. APEX is the first adaptive hand-controls.

Three files in. Coaching report out.

Sarah uploads her telemetry CSV, her FIA Certificate of Adaptations PDF, and a 30-second written debrief. The system processes the COA at the tensor level. It extracts the hand-control hardware specification + produces the COA-overview report. It then produces the coaching report every layer downstream.

Telemetry

1-Hz mini-sector tensors. Speed + acceleration + lateral G + RPM.

COA PDF

FIA Appendix L Certificate of Adaptations. Granite-Docling parse.

Debrief

Driver's 30-second written debrief. Granite Instru

Three layers between tensor and recommendation.

Layer 1: Forecast

Frozen Granite TimeSeries TTM r2.1 (NeurIPS 2024). Zero-shot multivariate. 1-Hz mini-sector telemetry to next-session pace envelope.

Layer 2: Project

Differentiable CvxpyLayer QP. Friction ellipse + for kinematic + jerk bound. SCP outer loop 3 iterations

8-TIER PHYSICS

SCP UNROLLED 3 ITERS

D-012 + D-013 + D-015

Three agents read every recommendation before the driver does.

Physics-Critic

Validates physics-projection feasibility per the Stage 1 + Stage 2 violation log.

Pedagogy-Critic

Validates coaching-report claims map back to the telemetry evidence.

If any critic flags, IBM Mellea Instruct-Validate-Repair fires with loop budget three until the panel approves audit. Three independent agents plus the safety gate.

Provenance on every line.

Corner-by-corner insights. Tuning recommendations. Forecast envelope. Every claim cites a specific FIA or specific COA section via the provenance footer.

The Watson TTS walkie-talkie reads the report in race-engineer voice with the paddock-radio filter chain (lowpass=3000 + compand + volume=1.8). Sarah hears it on her cool-down lap. Web Speech API fallback to FFmpeg (per D-043 Vercel architectural constraint).

FIA APPENDIX L

COA SECTION CITE

WATSON TTS

WALKIE-TALKIE FILTER

Sub-15-second coaching without a connection.

If Sarah loses connection in the paddock, the Granite 4.0 Nano 350M model runs in her browser via WebGPU. Same output shape. No round-trip.

350M

Granite 4.0 Nano parameters



Coaching-re

D-019 ITEM 1	D-021 IN-BROWSER INFERENCE	TRANSFORMERS.JS
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Stacked on top of the physics + critic + audit chain.

- **WebGPU Granite Nano 350M** — in-browser edge inference, no round-trip
- **Activated LoRA Layer 6** — vLLM hot-swap, KV-cache preserved, < 200 ms swap budget
- **GEPA reflective prompt optimization** — offline DSPy genetic-style faithfulness lift
- **EAGLE-3 speculative decoding** — draft + accept, 2.5-3.7x typical band per arXiv:2503.01840
- **Tri-agent Agent-as-Judge critic loop** — Physics + Pedagogy + Guardian-Safety

COA-parameterized brake-throttle simultaneity gate.

The same coaching pipeline produces different physics corrections for adaptive versus able-bodied drivers. The FIA Certificate of Adaptations.

When the COA permits simultaneous brake + throttle (as adaptive racing programmes + adapted-hand-controllers do), Stage 2's feasibility filter recognises it. When the COA does not permit it, the gate flags the input. To our knowledge, through 2026-Q2, the first public AI race-engineer workflow we found that reads the COA at the tensor

Twelve IBM tools. Every slot load-bearing.

Granite-Docling 258M

FIA COA PDF parser

Granite Vision 4.1 4B

Timing-sheet parser

Granite Tool r2.1

NeurIPS 2024

Granite Embedding R2

Hybrid dense + sparse RAG

IBM TSPulse 1M

Time-frequency anomaly

Granite 4

Race-engineer

Granite 4.0 Nano 350M

WebGPU edge

Langflow

Visible orchestration demo

Docling 1

Conversion + t

BEAT 8 • CLOSE

APEX

Built on IBM Granite.
For adaptive racers first,
every grassroots competitor second.

APEX DOT RACE

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APACHE 2.0